

Online Technical Test

DAC/PG-DAC Selection — 60+60 Format

Part A (60 min, AI Allowed)

Structure: Each set has 3 tiers in Part A . With 60 minutes each, candidates have real time to build meaningful work and demonstrate true understanding.

Scoring: - Part A (Build With AI) = 30 marks (Tier 1: 10, Tier 2: 10, Tier 3: 10)

Rule: Candidate MUST complete Tier 1 before Tier 2, Tier 2 before Tier 3. No skipping.

SET 1: Question Bank Browser

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Question Bank Table

Build an HTML page with a styled table showing 10 questions from a question bank.

- Create a JSON array: 10 questions with id, text, type (MCQ/FILL_BLANK), subject (HTML/CSS/JavaScript/SQL), difficulty (EASY/MEDIUM/HARD), status (DRAFT/APPROVED/REJECTED), marks
- Display in a clean table with all columns
- Status badges colour-coded: DRAFT = grey, APPROVED = green, REJECTED = red
- Difficulty colour-coded: EASY = blue, MEDIUM = orange, HARD = red
- Professional styling — this should look like a real admin panel

TIER 2 — Add Complexity (10 marks): Filtering + Search

- Add 3 filter dropdowns above the table: Subject, Difficulty, Status
- All 3 filters must work TOGETHER (combined filtering — not resetting each other)
- Add a real-time search box filtering by question text
- Show count: “Showing X of Y questions”

TIER 3 — Advanced Feature (10 marks): Pagination + Select/Delete + Sort

- Add pagination: 5 questions per page, Previous/Next buttons, “Page X of Y”
- Add checkboxes on each row + “Select All” checkbox in header
- “Delete Selected” button — removes selected questions, updates count and pagination
- Pagination must work correctly WITH active filters
- Clicking column headers sorts the table (ascending/descending toggle with ↑/↓ indicator)

SET 2: Test Configuration Form

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Test Configuration Form

Build a “Create New Test” form for an exam system.

- Fields: Title, Description (textarea), Subject (dropdown: HTML/CSS/JavaScript/SQL), Duration (minutes), Total Marks, Passing Marks, Difficulty (radio: EASY/MEDIUM/HARD)
- Checkboxes: Show Result, Show Certificate, Shuffle Questions
- Validation: Title required (min 5 chars), Duration 10–180, Passing Marks $\leq$ Total Marks
- On submit: show a styled “Preview Card” summarising all settings

TIER 2 — Add Complexity (10 marks): Date Validation + Duplicate

- Add Start Date and End Date pickers — End Date must be after Start Date (show error if not)
- Add Max Attempts field (number, default 1, max 5)
- Add a “Duplicate Test” button on the preview card that repopulates the form with all values (Title becomes “Copy of [original]”)
- Show character count under Description (e.g., “45/500 characters”)

TIER 3 — Advanced Feature (10 marks): Question Selection Panel

- Below the form, add a “Select Questions” section
- Show a mini table of 10 available questions (from JSON): Text (truncated), Subject, Difficulty, Marks
- Checkboxes to select questions for this test
- Auto-calculate and update Total Marks based on selected questions’ marks
- If selected questions’ total exceeds the entered Total Marks — show a warning
- Show count: “5 questions selected (Total: 25 marks)”
- Add a Subject filter dropdown to filter available questions

SET 3: Student Results Dashboard

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Results Page

Build a results page for a student who completed an exam.

- Create JSON: studentName, testTitle, totalMarks: 100, scoredMarks: 72, percentage: 72, passed: true, timeTaken: “42 min”
- Result Summary Card: student name, test, score (72/100), percentage, PASSED/FAILED badge (green/red), time taken
- Professional card styling with shadows and spacing

TIER 2 — Add Complexity (10 marks): Question-Wise Breakdown

- Add JSON array: 10 questionResults with questionText, selectedAnswer, correctAnswer, isCorrect, marks, subject
- Table below summary: Question #, Text, Your Answer, Correct Answer, ✓/✗, Marks
- Wrong answers = light red row, Correct = light green row
- Show sub-totals: “Correct: 7 | Wrong: 3 | Marks: 72/100”

TIER 3 — Advanced Feature (10 marks): Subject Analysis + Visuals

- Build a CSS-only circular or semi-circular progress indicator showing 72%
- Add “Subject-wise Analysis” section — group questions by subject dynamically
- Show mini table: Subject | Total Questions | Correct | Score | Percentage
- Highlight subjects below 50% in red — “Needs Improvement”
- Add a simple CSS horizontal bar chart showing per-subject performance



SET 4: SQL Schema & Queries for Question Bank

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Schema + Basic Queries

Write SQL for a question bank system.

- CREATE TABLE: users (id SERIAL PK, name, email UNIQUE, role TEXT), questions (id SERIAL PK, text, type, subject, difficulty, marks, review_status, created_by FK, created_at)
- INSERT 3 users, 10 questions
- Queries: (a) All approved questions sorted by subject, (b) Count of questions per subject
- Build a simple HTML page displaying 10 questions in a styled table

TIER 2 — Add Complexity (10 marks): Options Table + Joins

- CREATE TABLE: options (id, question_id FK, option_text, is_correct BOOLEAN, sort_order) with CASCADE delete
- INSERT 4 options per MCQ question (at least 5 questions)
- Queries: (c) All MCQ questions with their correct answer (JOIN), (d) Questions where NO option is marked correct (data integrity check), (e) Top contributor — who created the most questions?

TIER 3 — Advanced Feature (10 marks): Complex Analytics

- Queries: (f) Subject-wise difficulty distribution — cross-tab showing count of EASY/MEDIUM/HARD per subject
- Subjects where less than 50% of questions are APPROVED — show Subject, Total, Approved, Approval %
- Find duplicate questions — same text entered twice
- Questions still in DRAFT for more than 7 days

SET 5: MCQ Exam Interface

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Basic MCQ Interface

Build an exam-taking interface.

- JSON: 5 MCQ questions with text, 4 options each, correct answer
- Show one question at a time: “Question 1 of 5”
- 4 options as radio buttons — selected option highlighted
- “Next” button to move forward
- After last question, “Submit” button — show score: “You scored X out of 5”

TIER 2 — Add Complexity (10 marks): Navigation + Timer + Palette

- Add “Previous” button — user can go back and change answers
- Add a question palette on the side: buttons 1–5, colour-coded (grey = not visited, blue = answered, white = visited but not answered)
- Add a countdown timer (set to 10 minutes) at the top — updates every second

TIER 3 — Advanced Feature (10 marks): Review + Auto-Submit + Summary

- Add “Mark for Review” checkbox — marks the question yellow in the palette
- When timer reaches 0 — auto-submit the test (no user action needed)
- Before final submit, show a summary panel: Answered: X, Unanswered: Y, Marked for Review: Z
- Two buttons on summary: “Go Back to Review” and “Confirm Submit”
- After submit: show detailed results — each question, selected vs correct, right/wrong

SET 6: Certificate Generator

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): Certificate Template

Build a printable certificate of achievement.

- Landscape-oriented certificate (CSS width > height)
- Content: decorative border (CSS), organisation name “SPIPL Online Examination Portal”, “Certificate of Achievement”, student name, test name, score, date, certificate number (e.g., “CERT-2026-00123”)
- Two signature lines: “Examiner” and “Director”
- Professional styling — this should look like a real certificate

TIER 2 — Add Complexity (10 marks): Live Preview + Print

- Below the certificate: a form — Student Name, Test Name, Score, Date
- Certificate updates in real-time as the form is filled (live preview)
- “Print Certificate” button — opens browser print dialog
- Use `CSS@media print` to hide the form and show ONLY the certificate when printing

TIER 3 — Advanced Feature (10 marks): Verification + Templates

- Add a “Certificate Verification” section: input field for certificate number, “Verify” button
- If matches, show “Valid — Issued to [Name] for [Test] on [Date]” in green
- If not found, show “Invalid Certificate Number” in red
- Add a template switcher: 2 different certificate designs (“Classic” with border and “Modern” with gradient header). Dropdown switches between them — live preview updates
- Store generated certificates in a JavaScript array



SET 7: Admin User Management

PART A — Build With AI (60 minutes)

TIER 1 — Core Application (10 marks): User Table

Build an admin user management page.

- JSON: 12 users with id, name, email, role (ADMIN/CONTRIBUTOR/CANDIDATE), isActive (true/false), createdAt
- Styled table: Avatar circle (first letter of name), Name, Email, Role (coloured badge), Status (Active green / Inactive red), Created Date
- Role badges: ADMIN = purple, CONTRIBUTOR = blue, CANDIDATE = green

TIER 2 — Add Complexity (10 marks): Filters + Search + Modal

- Filter dropdowns: Role, Status (Active/Inactive/All)
- Search box filtering by name or email (real-time)
- “Add New User” button opens a modal with form: Name, Email, Role dropdown
- On modal submit: new user appears in the table
- Delete button on each row with confirmation: “Delete [name]?”

TIER 3 — Advanced Feature (10 marks): Pagination + Bulk Actions + Sort

- Pagination: 5 users per page, Previous/Next, “Page X of Y”
- Pagination works correctly WITH active filters
- Column header clicking sorts the table (ascending/descending toggle with ↑/↓)
- Checkboxes on each row + “Select All” for current page
- Bulk action dropdown: “Deactivate Selected”, “Delete Selected” — with confirmation